Practical 5 – Working with JSP and JSTL
Theory :-
What is JSP?
What are the elements of JSP?
Write about JSTL

5a. Create an html page with fields, eno, name, age, desg, salary. Now on submit this data to a JSP page which will update the employee table of database with matching eno.

```
create table emp(empid varchar(10) PRIMARY KEY, ename varchar(50), salary varchar(50),age varchar(50) );
insert into emp values('1','aaa','221234','11') ;
insert into emp values('2','bbb','334567','22') ;
insert into emp values('3','ccc','44454','33') ;
insert into emp values('4','ddd','55123','44');
```

Index5a.html

```
<html>
<body>
<form action="UpdateEmp.jsp" >
 Enter Employee Number<input type="text" name="txtEno" ><br>
 Enter Name<input type="text" name="txtName" ><br>
 Enter age<input type="text" name="txtAge" ><br>
 Enter Salary<input type="text" name="txtSal" ><br>
<input type="reset" ><input type="submit">
</form>
</body>
</html>
```

UpdateEmp.jsp

```
<%@page contentType="text/html" import="java.sql.*" %>
<html>
<body>
<h1>Updating Employee Record</h1>
<%
String eno=request.getParameter("txtEno");
String sal = request.getParameter("txtSal");
try{
Class.forName("com.mysql.jdbc.Driver");
Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb?autoReconnect=true&useSSL=false","root"
,"tiger");
PreparedStatement stmt = con.prepareStatement("select * from emp where empid=?");
stmt.setString(1, eno);
ResultSet rs = stmt.executeQuery();
if(rs.next()){
out.println("<h1> Employee "+rs.getString(2)+" Exist </h1>");
PreparedStatement pst= con.prepareStatement("update emp set salary=? where empid=?");
pst.setString(1, sal);
pst.setString(2, eno);
pst.executeUpdate();
out.println("<h1>Employee Record updated !!!!!</h1>");
}
else{
out.println("<h1>Employee Record not exist !!!!!</h1>");
}
}catch(Exception e){out.println(e);}
%>
```

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```
</body>
</html>
```

Q.5 b) Create a JSP application to demonstrate the use of JSTL.

Index5c.html

```
<html><body>
<a href="setDemo.jsp"> SetDemo</a>
<a href="Maxif.html"> MaxIF</a>
<a href="ForeachDemo.jsp"> ForEachDemo</a>
<a href="outDemo.jsp"> OutDemo</a>
<a href="URLDemo.jsp"> URLDemo</a>
<a href="choose_when_otherwise.jsp"> choose_when_otherwise</a>
</body></html>
```

setDemo.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jstl/core" %>
<c:set var="pageTitle" scope="application"
value="Dukes Soccer League: Registration" />
${pageTitle}
```

Maxif.html

```
<form action ="IFDemo.jsp">
 x=<input type="text" name="x" /><br>
 y=<input type="text" name="y" /><br>
 <input type="submit" value="Check Max" />
 </form>
```

IFDemo.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<c:set var="x" value="${param.x}"/>
<c:set var="y" value="${param.y}"/>
<c:if test="${x>y}">
 <font color="blue"><h2>The Ans is:</h2></font>
 <c:out value="${x} is greater than ${y}"/>
</c:if>
<c:if test="${x<y}">
 <font color="blue"><h2>The Ans is:</h2></font>
 <c:out value="${x} is Less than ${y}"/>
</c:if>
<c:if test="${x=y}">
 <font color="blue"><h2>The Ans is:</h2></font>
 <c:out value="${x} is equals ${y}"/>
</c:if>
```

ForeachDemo.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<c:forEach begin="1" end="10" var="i">
 The Square of <c:out value=" ${i}=${i*i}"/><br>
 </c:forEach>
```

outDemo.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<c:set var="name" value="John"/>
My name is: <c:out value= "${name}" />
```

URLDemo.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<c:url value="/index5c.html"/>
```

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choose_when_otherwise.jsp

```
<%@taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<c:set var="income" value="${4000*4}"/>
Your Income is: <c:out value="${income}"/>
<c:choose>
 <c:when test="${income <=1000}">
Income is not good
 </c:when>
 <c:when test="${income > 10000}">
 Income is Very Good
 </c:when>
 <c:otherwise>
 Income is undetermined
 </c:otherwise>
</c:choose>
```

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Practical 6 – Working with EJB
Theory –
What is EJB?
Types of EJB

a) Create a Currency Converter application using EJB.

Index.html

```
<html><head><title>Currency Converter</title></head>
<body>
<form action="CCServlet" >
Enter Amount <input type="text" name="amt"><br>
Select Conversion Type
<input type="radio" name="type" value="r2d" checked>Rupees to Dollar
<input type="radio" name="type" value="d2r" >Dollor to Rupees<br>
<input type="reset" ><input type="submit" value="CONVERT" >
</form>
</body>
</html>
```

CCServlet.java

```
package mypack;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.ejb.EJB;
import mybeans.CCBean;
import javax.servlet.annotation.WebServlet;
@WebServlet(urlPatterns = {"/CCServlet"})
public class CCServlet extends HttpServlet {
@EJB CCBean obj;
public void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
response.setContentType("text/html;charset=UTF-8");
PrintWriter out = response.getWriter();
double amt = Double.parseDouble(request.getParameter("amt"));
if(request.getParameter("type").equals("r2d"))
{
out.println("<h1>"+amt+ " Rupees = "+obj.r2Dollor(amt)+" Dollors</h1>");
}
if(request.getParameter("type").equals("d2r"))
{
out.println("<h1>"+amt+ " Dollors = "+obj.d2Rupees(amt)+" Rupees</h1>");
}
}}
```

CCBean.java

```
package mybeans;
import javax.ejb.Stateless;
@Stateless
public class CCBean {
public CCBean(){}
public double r2Dollor(double r){   return r/65.65;  }
public double d2Rupees(double d){   return d*65.65;  }
}
```

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b) Develop a Simple Room Reservation System Application Using EJB.

```
Create table roombook(RoomId varchar(4) PRIMARY KEY, RoomType varchar(20), charges numeric(10,2),
cust varchar(20), mob varchar(20) , status varchar(10)) ;
insert into roombook values('1001','Delux',5000.00,",",'Not Booked') ;
insert into roombook values('1002','Super Delux',7000.00,",",'Not Booked');
insert into roombook values('1003','Suit',9500.00,",",'Not Booked') ;
insert into roombook values('2001','Delux',5000.00,",",'Not Booked');
insert into roombook values('2002','Super Delux',7000.00,",",'Not Booked') ;
insert into roombook values('2003','Suit',9500.00,",",'Not Booked') ;
-----------------------------------RoomBook.html -----------------------------------
<form action='RBServlet' >
Select a room Type
<input type="radio" name="txtType" value="Delux">Delux
<input type="radio" name="txtType" value="Super Delux">Super Delux
<input type="radio" name="txtType" value="Suit">Suit<br>
Enter Your Name  <input type="text" name="txtCust" ><br>
Enter Mobile No. <input type="text" name="txtMob" ><br>
<input type="reset" ><input type="submit" value="BookRoom">
</form>
----------------------------------------RBServlet--------------------------------------
package mypack;
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.ejb.EJB;
import mybeans.RRBean;
import jakarta.servlet.annotation.WebServlet;
@WebServlet("/RBServlet")
public class RBServlet extends HttpServlet {
@EJB RRBean obj;
public void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException{
PrintWriter out=response.getWriter();
String rt=request.getParameter("txtType");
String cn=request.getParameter("txtCust");
String cm=request.getParameter("txtMob");
String msg = obj.roomBook(rt, cn, cm);
out.println("<html><body bgcolor=yellow text=red><h1>");
out.println(msg);
out.println("</h1></body></html>");
}}
----------------------------------------RRBean.java ----------------------------------
package mybeans;
import jakarta.ejb.Stateless;
import jakarta.ejb.LocalBean;
import java.sql.*;

@Stateless
@LocalBean
public class RRBean {
public RRBean(){}
public String roomBook(String rt, String cn, String cm){
String msg="";
try{
Class.forName("com.mysql.cj.jdbc.Driver");
Connection con;
```

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```
con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb?autoReconnect=true&useSSL=false","root"
,"tiger");
String query="select * from roombook where RoomType=? and status='Not Booked'";
PreparedStatement pst = con.prepareStatement(query);
pst.setString(1,rt);
ResultSet rs= pst.executeQuery();
if(rs.next()){
String rno=rs.getString(1);
PreparedStatement stm1 = con.prepareStatement("update roombook set cust=? where RoomId=? ");
PreparedStatement stm2 = con.prepareStatement("update roombook set mob=? where RoomId=? ");
PreparedStatement stm3 = con.prepareStatement("update roombook set status=? where RoomId=? ");
stm1.setString(1,cn); stm1.setString(2,rno);
stm2.setString(1,cm); stm2.setString(2,rno);
stm3.setString(1, "Booked"); stm3.setString(2,rno);
stm1.executeUpdate();
stm2.executeUpdate();
stm3.executeUpdate();
msg = "Room "+rno+ " Booked. The Charges are "+rs.getString(3);
}
else
{
msg = "Room "+rt+ " currently Not available";
}
}catch(Exception e){msg=""+e;}
return msg;}}
```

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